QUALIFICATION
REQUIREMENTS REMOVED

[INCH-POUND]

MIL-M-38510/136A
5 March 1990
SUPERSEDING
MIL-M-38510/136
6 December 1985

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, PRECISION VOLTAGE REFERENCES,
HYBRID SILICON

[Inactive for new design after date of this revision]

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for hybrid voltage references. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Classification.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
	Precision voltage reference;
01	+10.000 V
02	+10.000 V, -10.000 V

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-H-38534.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
C	D-1 (14-lead, .785" x .310" x .200"), dual-in-line package
X	(14-lead, .885" x .520" x .225"), dual-in-line package, (see figure 1)

1.3 Absolute maximum ratings.

Supply voltage (V_{IN}):	
Device type 01	±20 V dc
Device type 02	±20 V dc
Power dissipation:	
Device type 01	300 mW
Device type 02	450 mW
Lead temperature (soldering, 10 seconds)	+300°C
Storage temperature	-65°C to +150°C
Short circuit protection (to GND)	Continuous

[Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.]

AMSC N/A FSC 5962

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1.4 Recommended operating conditions.

Supply voltage range:	
Device type 01	+13.5 V dc to +16.5 V dc
Device type 02	±13.5 V dc to ±16.5 V dc
Output current (maximum)	5 mA 1/
Ambient operating temperature range (T_A)	-55°C to +125°C

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
14-lead dual-in-line package	C	300 mW at T_A = +125°C	50°C/W	120°C/W
14-lead dual-in-line package	X	500 mW at T_A = +125°C	20°C/W	60°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-H-38534 - Hybrid Microcircuits, General Specification for.
MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-H-38534, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-H-38534 and herein.

1/ With resistive load to common.

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3.2.1 Terminal connections. The terminal connnections shall be as specified on figure 2.

3.2.2 Functional schematic circuits. The functional schematic circuits shall be as shown on figure 3.

3.2.3 Case outlines. The case outlines shall be in accordance with 1.2.3 and on figure 1.

3.2.4 Package and sealing material. Package and sealing material shall be in accordance with MIL-H-38534.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-H-38534 (see 6.4).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I and apply over the full recommended ambient operating temperature range. Unless otherwise specified, the following conditions shall also apply.

Ambient temperature range: $-55^{\circ}C \leq T_A \leq +125^{\circ}C$
Device 01, $V_{IN} = +15$ V
Device 02, $V_{IN} = \pm 15$ V
$R_L = 2$ kΩ to common

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein. The "JAN" or "J" certification mark shall not be used.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number I (see MIL-M-38510, appendix E).

3.8 Manufacturer eligiblity. To be eligible to supply microcircuits to this specification a manufacturer shall have a manufacturer certification in accordance with MIL-H-38534 for at least one line, not necessarily the line producing the device type described herein.

3.9 Certification. Certification in accordance with MIL-H-38534 is not required for these devices.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions (see 3.4 and figure 5) unless otherwise specified 1/	Device type	Limits Min	Limits Max	Unit
Output voltage 2/	V_R	$T_A = +25^{\circ}C$	01	9.9975	10.0025	V
	V_R		02	9.9975	10.0025	V
	$-V_R$		02	-10.0025	-9.9975	V
3/	V_R	$-55^{\circ}C \leq T_A \leq +125^{\circ}C$	01	9.9945	10.0055	V
	V_R		02	9.9945	10.0055	V
	$-V_R$		02	-10.0055	-9.9945	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions (see 3.4 and figure 5) unless otherwise specified 1/	Device type	Limits Min	Limits Max	Unit
Quiescent current	I_{CC}	T_A = +25°C No load	01	.25	14	mA
			02	.25	17	mA
	I_{EE}		02	-4	-.25	mA
Line regulation	V_{RLINE}	13.5 V ≤ V_{IN} ≤ 16.5 V; T_A = +25°C	01,02	-300	+300	$\frac{\mu V}{V}$
Load regulation	V_{RLOAD}	0 mA ≤ I_L ≤ 10 mA; T_A = +25°C	01,02	-100	+100	$\frac{\mu V}{mA}$
Offset adjust range (Abs value)	V_{RADJ}	T_A = +25°C V_{RADJ+}	01	20	---	mV
		V_{RADJ-}	01	---	20	mV
	$+V_{RADJ}$	V_{RADJ+}	02	20	---	mV
		V_{RADJ-}	02	---	20	mV
	$-V_{RADJ}$	V_{RADJ+}	02	20	---	mV
		V_{RADJ-}	02	---	20	mV
Output short circuit current	I_{OS}	T_A = +25°C	01,02	---	60	mA
Output noise	N_O	V_O = 10 V; T_A = +25°C; See figure 6 no load; 0.1 Hz ≤ BW ≤ 10 Hz	01,02	---	50	μVp-p

1/ Tested with no offset adjustment.

2/ Allow 3 minutes after power-on before testing, T_A = +25°C (±2°C).

3/ Output voltage change as a function of temperature is determined using the box method. Each device is tested at -55°C, +25°C, and +125°C. At each temperature the output voltage (V_{OUT}) must fall within the rectangular area bounded by the minimum and maximum temperature. This method gives a maximum temperature coefficient of 9 ppm/°C and a typical value of 3 ppm/°C.

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TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5008)	1	1
Final electrical test parameters (method 5008) 1/	1, 2, 3	1, 2, 3
Group A test requirements (method 5008)	1, 2, 3, 4	1, 2, 3, 4
Group C end-point and group B, class S, electrical parameters (method 5008)	1, 2, 3 and table IV limits	1 and table IV limits

1/ PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38534 and method 5008 of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5008 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) Class S devices: Burn-in and steady state life test circuit (test condition B) using the circuit on figure 4. Test duration shall be 240 hours minimum. Static tests (test condition A) using the circuit shown on figure 4. Test duration shall be 48 hours minimum.

(2) Class B devices: Test condition B using the circuit shown on figure 4; or test condition D, using the circuit shown on figure 4; or test condition F, using the circuit shown on figure 4.

b. Interim and final electrical test parameters shall be as specified in table II except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

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c. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38534, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5008 of MIL-STD-883 and with no intervening electrical measurements. If initial electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If initial electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

d. Internal visual inspection method 2017 (hybrid and multichip criteria) of MIL-STD-883.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38534 and as specified herein. Inspections to be performed shall be those specified in method 5008 of MIL-STD-883, and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table X of method 5008 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 5, 6, 7, 8, 9, 10, and 11 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table XI of method 5008 of MIL-STD-883 and as follows:

a. Electrical parameters shall be as specified in table II herein. For class S devices, delta limits shall apply only to subgroup 5 of group B inspection.

b. Steady-state life test for class S devices shall be in accordance with table XI of method 5008 of MIL-STD-883, using the circuit shown on figure 4. If the alternate burn-in conditions are used, the circuit shown on figure 4 shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table XII of method 5008 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

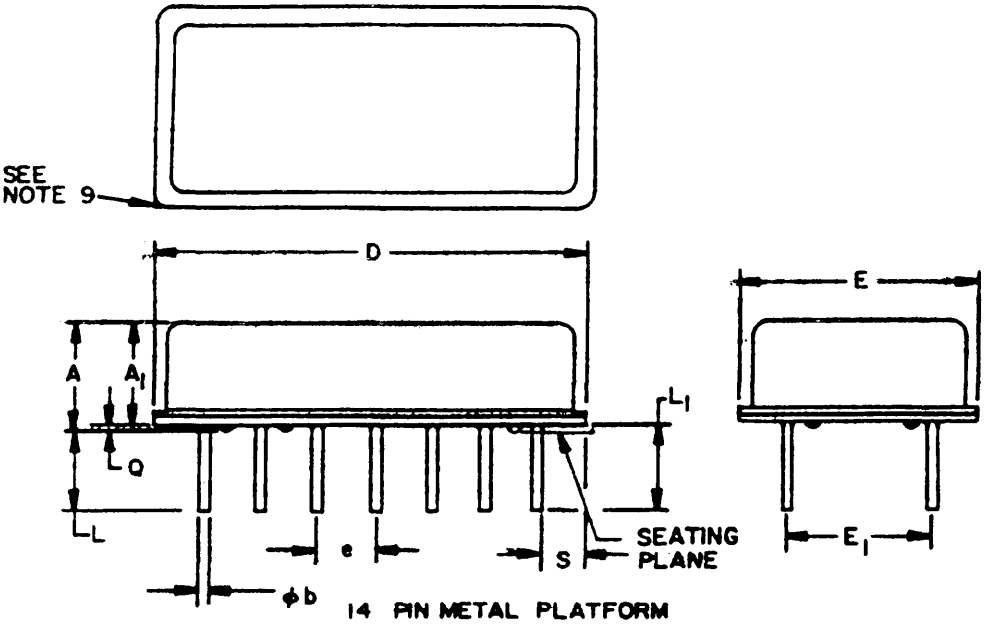
b. Steady-state life test (method 5008 of MIL-STD-883) conditions:

(1) Test condition B using the circuit shown on figure 4.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

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14 PIN METAL PLATFORM

Dimensions					
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.215	.225	5.46	5.72	
A_1	.175	.215	4.45	5.46	
φb	.016	.019	0.41	0.48	6
D	.860	.885	21.84	22.48	6
E	.490	.520	12.45	13.21	9
E_1	.295	.305	7.49	7.75	
e	.100 BSC		2.54 BSC		7, 11
L	.185	.255	4.70	6.48	
L_1	.205	.290	5.21	7.37	
Q	.020	.035	0.51	0.89	5
S	.100	.16	2.54	4.1	
S_1	.08	.18	2.0	4.6	

FIGURE 1. Case outline X.

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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Index area: A notch, square-package corner, or a pin one index point shall be located adjacent to pin one and within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
4. Dimension Q shall be measured from the seating plane to the base plane.
5. This dimension allows for off-center lid, meniscus and weld squash.
6. The basic pin spacing is .100 (2.54 mm) between centerlines. Each pin centerline shall be located within ±0.10 inch (0.25 mm) of its exact longitudinal position relative to pins 1 and 14.
7. Dimension S_1 is not used.
8. Lead center when a = 0°. E_1 shall be measured at the centerline of the leads (see appendix C of MIL-M-38510).
9. All leads - increase maximum limit by 0.003 inch (0.08 mm) measured at the widest diameter when lead finish A or B is applied.
10. Twelve places.
11. If this configuration is used, no polymer or organic materials shall be applied or molded to the bottom of the package or cover the leads.

FIGURE 1. Case outline X - Continued.

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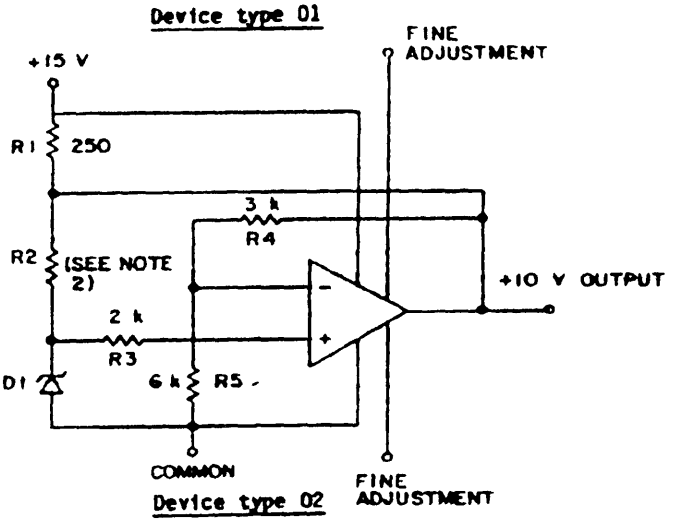
Device types	01	02
Case outlines	C and X	
Terminal number	Terminal symbol	
1	NC	- Fine adjust
2	NC	- V_{OUT} (-10 V)
3	NC	- Fine adjust
4	NC	- V_{IN} (-15 V)
5	NC	NC
6	NC	NC
7	Common	Common
8	NC	NC
9	NC	NC
10	Test point	Test point
11	+V_{IN} (+15 V)	+V_{IN} (+15 V)
12	Fine adjust	+Fine adjust
13	+V_{OUT} (+10 V)	+V_{OUT} (+10 V)
14	Fine adjust	+Fine adjust

NC = No connection

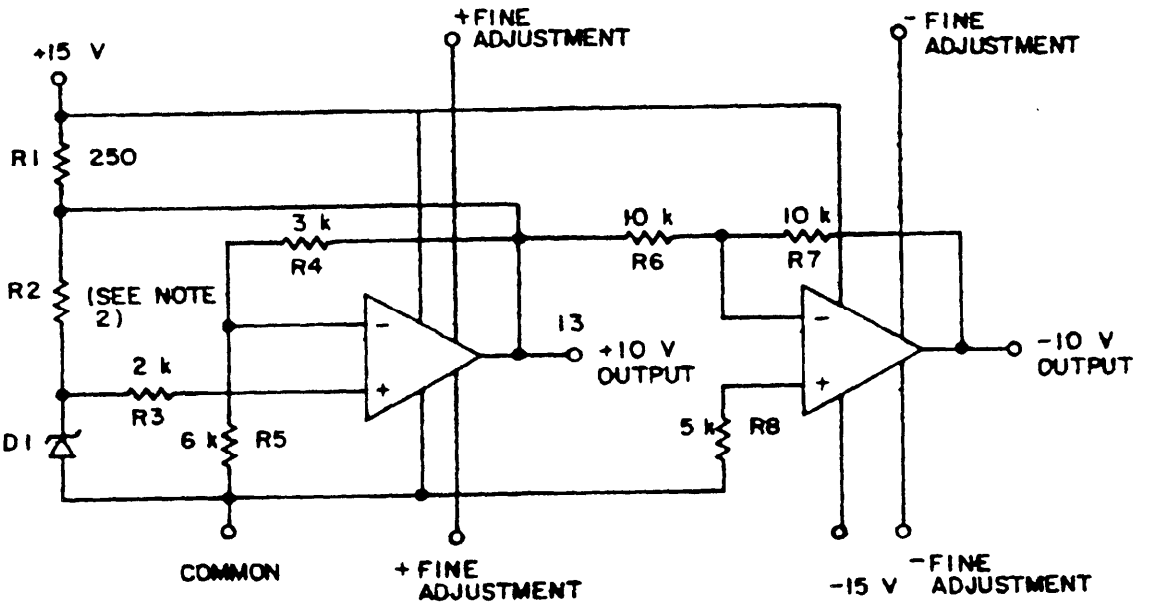
FIGURE 2. Terminal connections.

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Device type 01



Device type 02

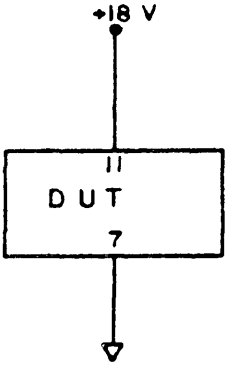


NOTES:
1. All resistor values are nominal and in ohms.
2. The value of R2 may vary between 100 to 10Ω depending on the value of the zener diode, D1.

FIGURE 3. Functional schematic circuit.

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Device type 01



Device type 02

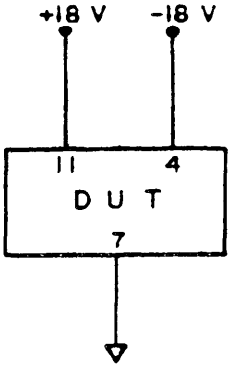
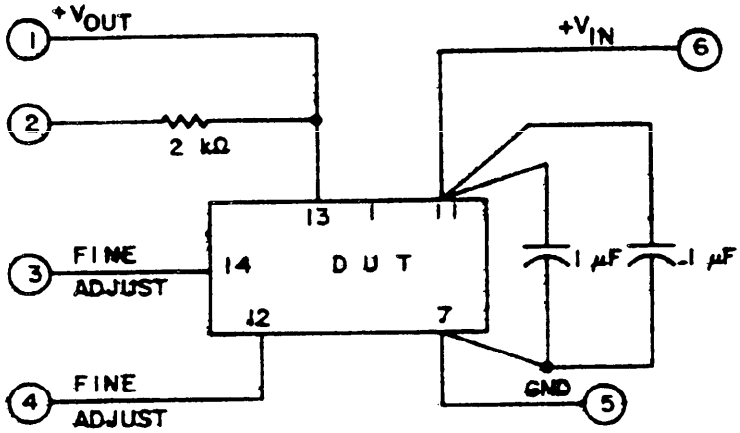


FIGURE 4. Test circuit for burn-in and steady state life test.

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Device type 01



Device type 02

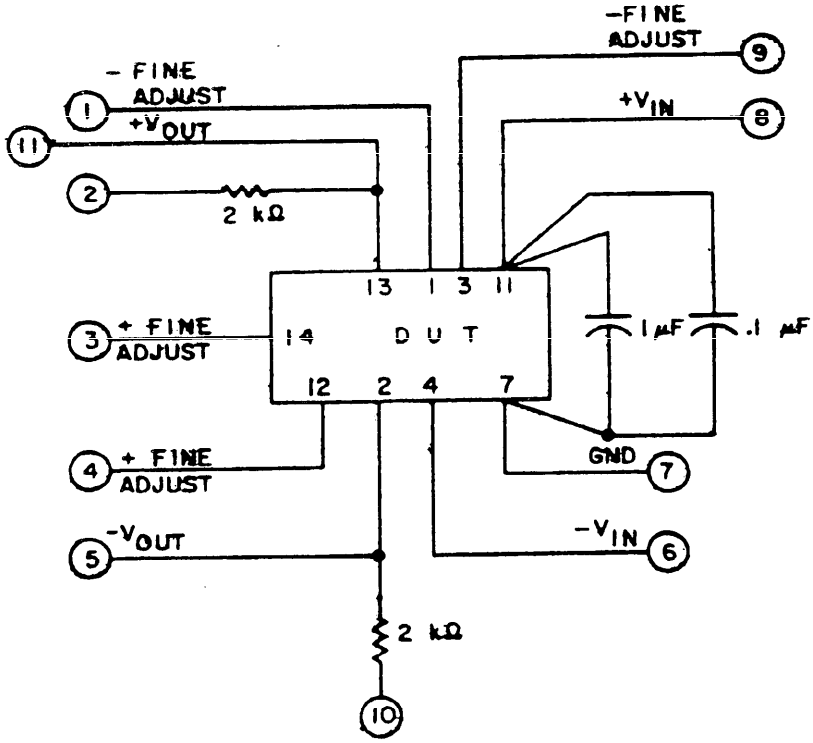
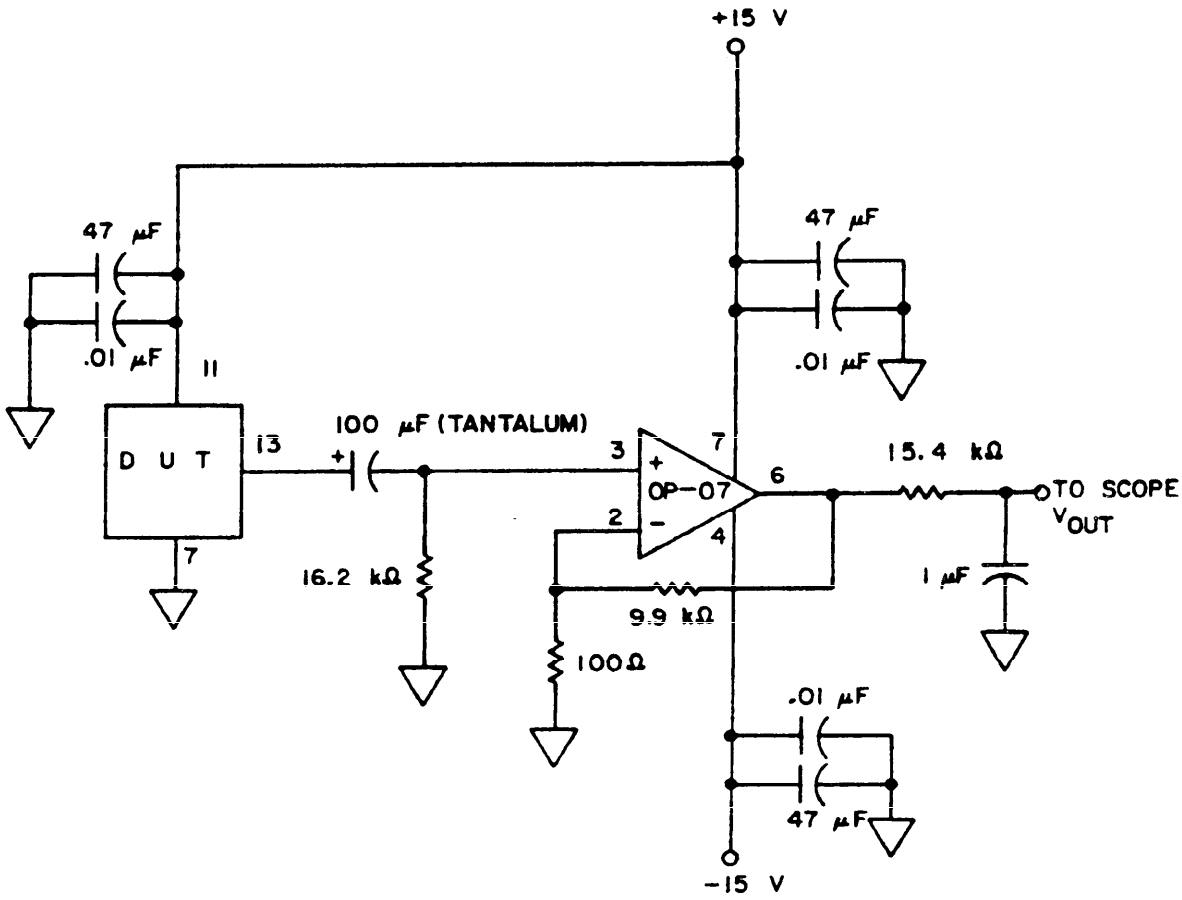


FIGURE 5. Test circuit for static tests.

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Device type 01

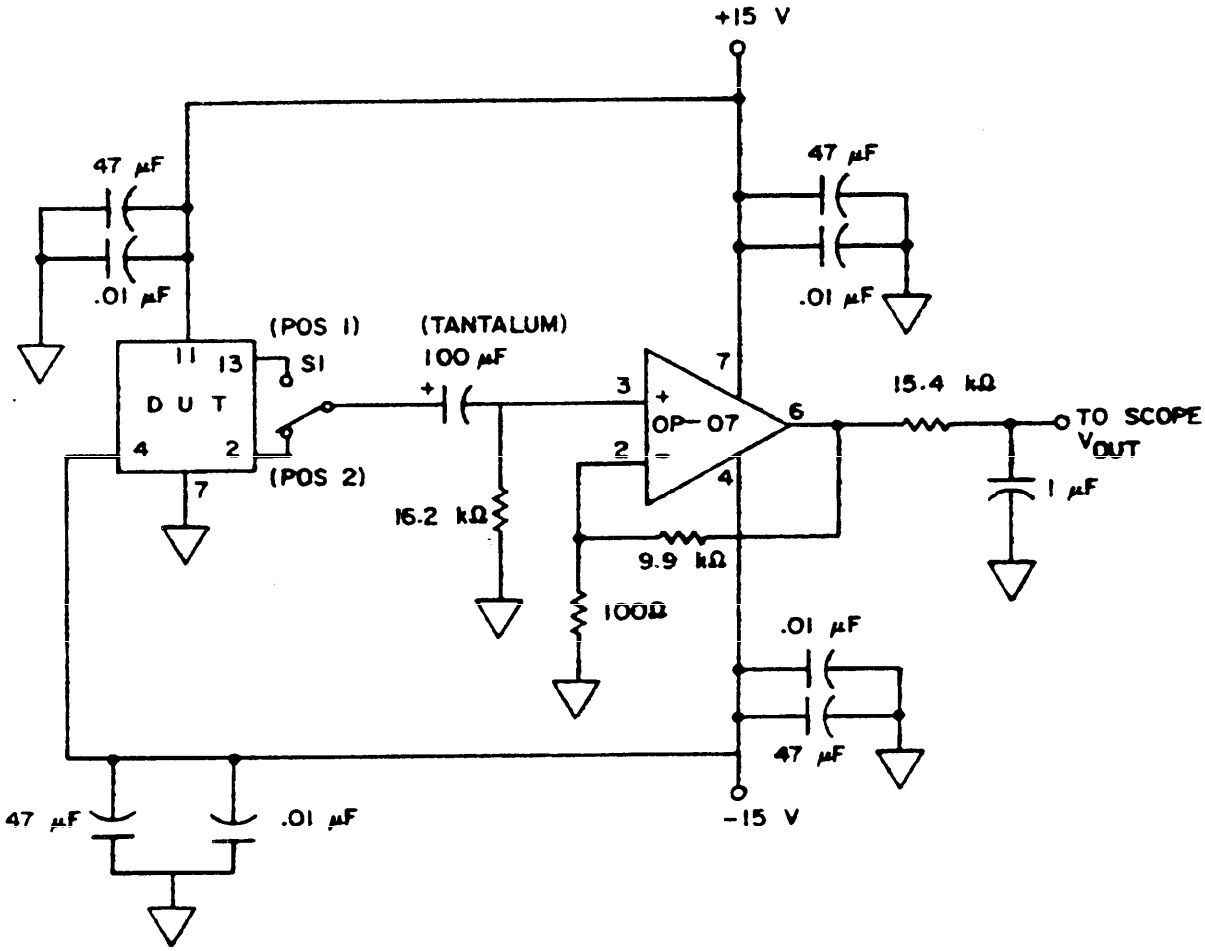


NOTES:
1. Test time = 10 seconds.
2. V_{OUT} measured with differential amplifier 7A22 and lower frequency set to 0.1 Hz.

FIGURE 6. Low frequency noise test circuit.

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Device type 02



NOTES:
1. Test time = 10 seconds.
2. V_{OUT} measured with differential amplifier 7A22 and lower frequency set to 0.1 Hz.
3. V_{OUT} = 10 V, S_1 position 1, V_{OUT} = -10 V, S_1 position 2.

FIGURE 6. Low frequency noise test circuit - Continued.

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TABLE III. Group A inspection for device type 01.

Subgroup	Symbol	Test no.	Test conditions (see figure 5) applied voltage and current on test pins							Measured pin			Equation	Limits		Unit
			1	2	3	4	5	6		No.	Value	Units		Min	Max	
1 T_A = +25°C	V_R	1		GND			GND	15 V		1	E_1	V	$V_R = E_1$	9.9975	10.0025	V
	I_{CC}	2					"	15 V		6	I_1	mA	$I_{CC} = I_1$	.25	14	mA
	V_{RLINE}	3 4		GND GND			" "	13.5 V 16.5 V		1 1	E_2 E_3	V V	$V_{RLINE} = (E_3 - E_2)\ 1000000/3$	-300	+300	$\frac{\mu V}{V}$
	V_{RLOAD}	5 6	0 mA 10 mA					15 V "		1 1	E_4 E_5	V V	$V_{RLOAD} = (E_4 - E_5)\ 100,000$	-100	+100	$\frac{\mu V}{mA}$
	$+V_RADJ$	7		GND		GND	GND	"		1	E_6	V	$+V_RADJ = E_6 - E_1$	20	---	mA
	$-V_RADJ$	8		"	GND		"	"		1	E_7	V	$-V_RADJ = E_7 - E_1$	---	20	mV
2 T_A = +125°C	V_R	9		"			"	"		1	E_8	V	$V_R = E_8$	9.9945	10.0055	V
3 T_A = -55°C	V_R	10		"			"	"		1	E_9	V	$V_R = E_9$	9.9945	10.0055	V
4 T_A = +25°C	I_{OS}	11	GND				"	"		1	I_2	mA	$I_{OS} = I_2$	---	60	mA
	N_O	12					"	"	(see figure 6)	---	E_{10}	---	$N_O = E_{10}$	---	50	$\mu Vp-p$

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TABLE III. Group A inspection for device type 02 - Continued.

Subgroup	Symbol	Test no.	Test conditions (see figure 5) applied voltage and current on test pins											Measured pin			Equation	Limits		Unit
			1	2	3	4	5	6	7	8	9	10	11	No.	Value	Units		Min	Max	
1 T_A = +25°C	V_R	1		GND				-15 V	GND	15 V		GND		11	E_1	V	$V_R = E_1$	9.9975	10.0025	V
	$-V_R$	2		"				"	"	"		"		5	E_2	V	$-V_R = E_2$	-10.0045	-9.9948	V
	I_{CC}	3						"	"	"				8	I_1	mA	$I_{CC} = I_1$	.25	17	mA
	I_{EE}	4						"	"	"				6	I_2	mA	$I_{EE} = I_2$	-4.0	-0.25	µA
	V_{RLINE}	5		GND				"	"	13.5 V		GND		11	E_3	V	$V_{RLINE} = (E_4 - E_3)\frac{1000000}{3}$	-300	+300	±V
		6		"				"	"	16.5 V		"		16	E_4	V				
		7		"				-13.5 V	"	15 V		"		5	E_5	V	$V_{RLINE} = (E_6 - E_5)\frac{1000000}{3}$	-300	+300	±V
		8		"				-16.5 V	"	"		"		5	E_6	V				
	V_{RLOAD}	9						-15 V	"	"			0 mA	11	E_7	V	$V_{RLOAD} = (E_8 - E_7)\ 100,000$	-100	+100	±V
		10						"	"	"			10 mA	11	E_8	V				
		11					0 mA	"	"	"				5	E_9	V	$V_{RLOAD} = (E_{10} - E_9)\ 100,000$	-100	+100	±V
		12					10 mA	"	"	"				5	E_{10}	V				
	$+V_RADJ$	13		GND		GND		"	"	"		GND		11	E_{11}	V	$+V_RADJ = E_{11} - E_1$	20	---	mV
	$+V_RADJ$	14		"	GND			"	"	"		GND		11	E_{12}	V	$-V_RADJ = E_{12} - E_1$	---	20	mV
	$-V_RADJ$	15		"				"	"	"	-15 V	"		5	E_{13}	V	$-V_RADJ = E_{13} - E_1$	20	---	mV
	$-V_RADJ$	16	-15 V	"				"	"	"		"		5	E_{14}	V	$-V_RADJ = E_{14} - E_1$	---	20	mV
2 T_A = +125°C	V_R	17		"				"	"	"		"		11	E_{15}	V	$V_R = E_{15}$	9.9945	10.0055	V
	$-V_R$	18		"				"	"	"		"		5	E_{16}	V	$-V_R = E_{16}$	-10.0045	-9.9945	V
3 T_A = -55°C	V_R	19		"				"	"	"		"		11	E_{17}	V	$V_R = E_{17}$	9.9945	10.0055	V
	$-V_R$	20		"				"	"	"		"		5	E_{18}	V	$-V_R = E_{18}$	-10.0045	-9.0045	V
4 T_A = +25°C	I_{OS}	21		"				"	"	"		"		11	I_3	mA	$I_{OS} = I_3$	---	60	mA
		22		"			GND	"	"	"		"	GND	5	I_4	mA	$I_{OS} = I_4$	-60	---	mA
	N_O	23						"	"	"				---	E_{19}	---	$N_O = E_{19}$ (see figure 6)	---	50	µVp-p
		24												---	E_{20}	---	$N_O = E_{20}$	---	50	µVp-p

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table XIII of method 5008 of MIL-STD-883.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltages given are referenced to the pin 7 ground terminal of the device under test (DUT). Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and steady-state life test cooldown procedures. When the devices are measured at +25°C, following application of the steady-state life or burn-in test condition, they shall be cooled to within +10°C of their power stable condition prior to removal of the bias.

4.6 Data reporting. When specified in the contract or purchase order, a copy of the following data, as applicable, shall be applied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and operating life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameter data (see 4.2b).

TABLE IV. Group C end-point electrical parameters.

Device type	Test		T_A = +25°C		Limits		Unit
			Symbol	Delta limits 1/	Min	Max	
01 02	Output voltage	V_R = 10 V V_R = 10 V $-V_R$ = -10 V	V_{OUT} V_{OUT} V_{OUT}	±350 ppm ±350 ppm ±350 ppm	9.99 9.99 -10.01	10.01 10.01 -9.99	V V V

1/ Delta limits apply to the measured value (see delta limit definition in MIL-H-38534).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-H-38534.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.2 Acquisition requirements. The acquisition document must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Part or Identifying Number (PIN) (see 6.6).

d. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

e. Requirements for certificate of compliance, if applicable.

f. Requirements for notification of change of product to the contracting activity in addition to notification to the qualifying activity, if applicable.

g. Requirements for failure analysis (including required test condition method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

h. Requirements for product assurance options.

i. Requirements for special lead lengths, or lead forming, if applicable. These requirements shall not affect the part number.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-H-38534, MIL-STD-1331, and as follows:

6.3.1 Line regulation (V_{RLINE}). Line regulation is the change in the output voltage for a specified change in the input voltage.

6.3.2 Lead regulation (V_{RLOAD}). Load regulation is the change in the output voltage for a specified change in the load current.

6.3.3 Output noise (V_N). Output noise is the electrical noise, measured at the output, with a constant load current and a constant input voltage. The "1/f" noise is the peak-to-peak noise voltage measured over a frequency band from 0.1 Hz to 10 Hz.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510 and MIL-H-38534.

Military device type	Generic-industry type
01	2700, R675B-4
02	2702, R675B-3

6.6 Part or Identifying Number (PIN). The PIN shall be in accordance with MIL-M-38510, and as specified herein.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1173-2)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE	
MIL-M-38510/136 A	Microcircuits, Linear, Precision	Voltage References, Hybrid Silicon

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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of the specimen fully exposed. The apparatus shall be capable